How to Prove It - Example 2.3.3

Solutions in brief

Part 1

Investigate the logical form of $x \in \mathcal{P}\, A$.

Solution

$$x \in \mathcal{P}\, A \equiv (\forall z \mid z \in x : z \in A)$$

♥

Part 2

Investigate the logical form of $\mathcal{P}\, A \subseteq \mathcal{P} B$.

Solution

$$\mathcal{P}\, A \subseteq \mathcal{P} B \equiv (\forall X \mid (\forall z \mid z \in X : z \in A) : (\forall z \mid z \in X : z \in B))$$

♥

Part 3

Invesitgate the logical form of $B \in \{\mathcal{P}\, A \mid A \in \mathcal{F}\}$

Solution

$$B \in \{\mathcal{P}\, A \mid A \in \mathcal{F}\} \equiv (\exists A \mid (\forall z \mid z \in A : z \in \mathcal{F}) : (\forall z \mid : z \in B \equiv (\forall y \mid y \in z : y \in A)))$$

♥

Detailed answers

Each solution was discovered by direct calculation. I used the laws from LADM and organised the logical transformations using the linear proof layout of Back's refinement calculus.

Parts 1 and 2 were straightforward requiring only a couple of definitions to transform the given statement into a quantified term.

Calculation for Part 1.

◆	$x \in \mathcal{P}\, A$
$\equiv$	HTPI (2.3.2) Powerset
	$x \in \{x \mid x \subseteq A\}$
$\equiv$	LADM (11.6) Characteristic predicate
	$x \subseteq A$
$\equiv$	LADM (11.13) Axiom, Subset
	$(\forall z \mid z \in x : z \in A)$
■	

Calculation for Part 2.

$$
\begin{array}{ll}
\blacklozenge & \mathcal{P}\,A \subseteq \mathcal{P}B \\
\equiv & \text{LADM (11.13) Axiom, Subset} \\
 & (\forall X \mid X \subseteq A : X \subseteq B) \\
\equiv & \text{LADM (11.13) Axiom, Subset, twice} \\
 & (\forall X \mid (\forall z \mid z \in X : z \in A) : (\forall z \mid z \in X : z \in B)) \\
\blacksquare &
\end{array}
$$

Part 3 was interesting for two reasons. Firstly, I needed to reinterpret traditional notation for set-comprehension in the uniform syntax of LADM. Secondly, I used a sub-calculation to focus on manipulating a subterm, thus demonstrating a streamlined calculation.

Calculation for Part 3.

$$
\begin{array}{lll}
\blacklozenge & B \in \{\mathcal{P}\,A \mid A \in \mathcal{F}\} & \\
\equiv & \text{Interpret using LADM notation} & \\
 & B \in \{A \mid A \in \mathcal{F} : \mathcal{P}\,A\} & \\
\equiv & \text{LADM(11.3) Axiom, Set membership} & \\
 & (\exists A \mid A \in \mathcal{F} : B = \mathcal{P}\,A) & \\
\equiv & \text{Part 1.} & \\
 & (\exists A \mid (\forall z \mid z \in A : z \in \mathcal{F}) : B = \mathcal{P}\,A) & \\
\equiv & \text{Focus on term of quantification} & \\
 & \begin{array}{ll}
 \blacklozenge & B = \mathcal{P}\,A \\
 \equiv & \text{LADM (11.4) Axiom, Extensionality} \\
 & (\forall z \mid : z \in B \equiv z \in \mathcal{P}\,A) \\
 \equiv & \text{Part 1.} \\
 & (\forall z \mid : z \in B \equiv (\forall y \mid y \in z : y \in A)) \\
 \blacksquare &
 \end{array} & \\
\ldots & (\exists A \mid (\forall z \mid z \in A : z \in \mathcal{F}) : (\forall z \mid : z \in B \equiv (\forall y \mid y \in z : y \in A))) & \\
\blacksquare & &
\end{array}
$$

I delight in turning quanitifer-heavy formulae into pointfree expressions of relation algebra. This exercise is asking us to do the opposite of that. Yuck.